party libraries that provide TypeScript typings (e.g., `@types/react`), writing reusable and type-safe components for design systems, and production software.

A key resource supporting this ecosystem is the DefinitelyTyped repository (https://github.com/DefinitelyTyped/DefinitelyTyped), which hosts high-quality, community-maintained TypeScript type definitions for thousands of popular JavaScript libraries. These type definitions, published as `@types` packages on npm, enable TypeScript projects to use JavaScript libraries with full type safety and IntelliSense (IDE code-completion and editing) support, even if the original libraries do not include their own type declarations. This greatly enhances developer experience and code reliability when integrating third-party tools.

By leveraging DefinitelyTyped typings and IntelliSense, developers can confidently use a vast range of JavaScript packages in TypeScript projects, ensuring better maintainability, improved productivity, and fewer runtime errors.

3.7 Type system—type annotations, type inference, and type safety

At the heart of TypeScript lies its powerful type system, which enables developers to write maintainable and scalable code. Mastering TypeScript means not only understanding its basic building blocks—like type annotations and type inference—but also applying type-safe patterns in real-world applications and architecting advanced solutions for complex systems.

In technical interviews, questions around TypeScript's type system often go beyond syntax, testing candidates' ability to model APIs, enforce contracts, and reason about type safety under edge cases. Whether defining a simple interface or crafting intricate generic utilities, a strong grasp of these concepts is essential for modern frontend and backend development.

Table 3.5 serves as Jane's review guide for the key features of TypeScript's type system. Use it as a checklist to solidify your understanding before heading into interviews.